

WATERVILLE LAFLEUR, ME, USA

WMO: 726073

Lat: 44.533N

Lon: 69.667W

Elev: 310

StdP: 14.53

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14615

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	0.1	-20.1	1.8	-0.2	-15.1	2.4	3.5	24.6	21.5	21.5	23.4	4.9	290	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	86.0	71.4	82.4	69.4	80.7	67.8	73.7	82.7	71.7	80.0	69.9	76.8	7.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.9	115.2	78.7	69.6	110.0	76.8	66.5	98.6	73.9	37.4	83.0	35.6	79.8	34.0	77.6	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.0	17.5	15.3	-11.3	91.0	6.1	1.8	-15.7	92.3	-19.3	93.4	-22.8	94.4	-27.2	95.7	
			DB												
			WB	-12.0	77.3	5.3	2.2	-15.8	78.9	-19.0	80.3	-21.9	81.5	-25.8	83.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	19.2	22.3	31.4	43.2	54.6	62.9	68.7	67.3	59.9	48.2	36.9	25.8
	DBStd	18.89	11.37	10.33	9.35	7.41	6.56	6.28	4.95	4.96	7.04	7.42	8.57	9.54
	HDD50	3847	956	776	579	228	26	1	0	0	5	125	401	750
	HDD65	7566	1421	1196	1040	653	329	113	20	31	183	523	843	1214
	CDD50	2081	0	0	3	25	169	387	581	537	303	68	8	0
	CDD65	327	0	0	0	0	7	48	136	104	31	1	0	0
	CDH74	2625	0	0	6	12	138	443	1034	749	234	9	0	0
	CDH80	633	0	0	1	2	37	105	254	175	59	0	0	0
Wind	WSAvg	5.9	6.4	6.8	7.5	7.3	5.8	4.9	4.4	4.2	4.8	5.8	6.3	6.4
Precipitation	PrecAvg	43.0	3.1	2.5	3.6	3.7	3.3	3.9	3.5	3.1	3.9	4.5	4.2	3.9
	PrecMax	61.7	5.9	5.5	6.3	8.5	7.6	8.0	6.5	7.6	10.0	13.2	6.7	6.9
	PrecMin	26.4	0.5	0.5	1.2	0.4	0.4	1.4	1.7	0.4	1.0	0.8	1.6	1.1
	PrecStd	8.0	1.5	1.1	1.2	2.0	1.7	1.9	1.5	1.8	2.1	3.0	1.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	50.1	64.0	73.5	84.4	88.6	90.1	89.5	87.7	74.8	63.4	53.9
		MCWB	50.5	44.3	52.3	56.8	65.7	71.7	75.3	72.8	72.0	63.2	56.9	50.2
	2%	DB	45.0	45.2	53.9	65.9	78.5	82.5	85.7	84.2	80.7	69.1	58.6	47.8
		MCWB	41.8	40.0	44.9	52.2	61.0	68.6	72.3	70.7	68.5	59.4	53.8	44.9
	5%	DB	39.1	40.5	47.9	61.0	72.5	79.3	82.2	81.4	75.3	63.9	54.5	43.3
		MCWB	35.5	35.8	40.5	48.8	58.3	65.2	70.0	68.9	65.3	56.9	50.1	38.8
	10%	DB	35.6	36.8	44.7	55.5	68.1	75.2	80.5	78.7	72.1	60.9	49.9	38.6
		MCWB	32.0	32.8	37.9	45.6	55.4	63.5	68.4	66.8	63.3	54.3	45.3	34.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	46.1	54.3	60.3	68.6	74.5	77.2	75.4	73.7	66.2	59.4	51.4
		MCDB	52.0	48.0	61.6	68.9	80.3	84.9	86.5	85.0	84.3	70.6	62.4	52.5
	2%	WB	42.4	41.1	46.4	55.4	64.1	71.1	74.2	72.9	70.3	62.3	54.7	45.6
		MCDB	44.4	44.7	50.4	62.0	73.5	80.0	82.6	81.1	77.3	66.6	57.1	46.5
	5%	WB	36.2	36.9	41.4	50.9	60.8	68.5	72.2	70.8	67.6	58.7	50.5	39.4
		MCDB	39.3	40.0	46.9	57.5	69.4	75.8	80.2	78.3	73.2	63.5	54.0	42.5
	10%	WB	32.1	33.0	38.1	47.1	57.7	65.9	70.2	68.8	65.1	54.6	46.0	35.3
		MCDB	34.9	36.5	43.0	55.1	65.5	72.5	77.0	75.6	70.1	59.1	48.8	38.1
Mean Daily Temperature Range	5% DB	MDBR	16.5	17.3	16.7	18.3	20.2	19.2	19.1	19.4	19.3	17.4	14.3	14.9
		MCDBR	18.3	19.0	22.1	27.2	28.8	25.3	22.7	23.1	23.3	22.2	18.7	18.7
		MCWBR	17.3	15.6	15.0	16.0	15.8	14.0	11.9	12.1	13.7	14.7	15.6	16.8
	5% WB	MCDBR	18.8	18.1	20.2	23.8	24.7	21.3	19.9	20.0	20.2	20.2	17.4	18.4
		MCWBR	18.4	15.8	15.2	16.4	15.3	13.5	11.5	11.8	13.1	15.5	16.0	17.4
Clear-Sky Solar Irradiance	taub	0.272	0.287	0.320	0.349	0.372	0.415	0.429	0.402	0.363	0.328	0.311	0.282	
	taud	2.419	2.354	2.398	2.407	2.387	2.270	2.242	2.317	2.413	2.515	2.521	2.476	
	Ebn at Noon	273	287	291	290	285	272	266	269	272	270	256	258	
	Edh at Noon	23	29	32	35	37	42	43	38	32	25	21	20	
All-Sky Solar Radiation	RadAvg	520	798	1179	1432	1607	1738	1801	1633	1324	830	530	398	
	RadStd	40	67	68	159	212	146	146	94	81	76	48	36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	N/A

Nomenclature: See separate page